HYDROGEN CHLORIDE **ICSC: 0163 (November 2016)**
Anhydrous hydrogen chloride
Hydrochloric acid, anhydrous

CAS #: 7647-01-0
UN #: 1050
EC Number: 231-595-7

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible.		In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep cylinder cool by spraying with water. Combat fire from a sheltered position.

AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!			
	SYMPTOMS	**PREVENTION**	**FIRST AID**
Inhalation	Cough. Sore throat. Burning sensation. Shortness of breath. Laboured breathing.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Artificial respiration may be needed. Refer immediately for medical attention.
Skin	Redness. Pain. Serious skin burns. ON CONTACT WITH LIQUID: FROSTBITE.	Cold-insulating gloves. Protective clothing.	Wear protective gloves when administering first aid. First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Refer immediately for medical attention.
Eyes	Redness. Pain. Blurred vision. Severe burns. ON CONTACT WITH LIQUID: FROSTBITE.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion			

SPILLAGE DISPOSAL

Evacuate danger area! Consult an expert! Personal protection: gas-tight chemical protection suit including self-contained breathing apparatus. Ventilation. Remove gas with fine water spray.

STORAGE

Cool. Fireproof if in building. Separated from food and feedstuffs and incompatible materials. See Chemical Dangers. Keep in a well-ventilated room.

PACKAGING

CLASSIFICATION & LABELLING

According to UN GHS Criteria

DANGER
Contains gas under pressure; may explode if heated
Toxic if inhaled
Causes severe skin burns and eye damage
May cause respiratory irritation
See Notes

Transportation
UN Classification
UN Hazard Class: 2.3; UN Subsidiary Risks: 8

Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.

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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance
COLOURLESS COMPRESSED LIQUEFIED GAS WITH PUNGENT ODOUR.

Physical dangers
The gas is heavier than air and may accumulate in lowered spaces causing a deficiency of oxygen.

Chemical dangers
The solution in water is a strong acid. It reacts violently with bases and is corrosive. Reacts violently with oxidants. This produces toxic gas (chlorine - see ICSC 0126). Attacks many metals in the presence of water. This produces flammable/explosive gas (hydrogen - see ICSC 0001).

Formula: HCl
Molecular mass: 36.5
Boiling point: -85.1°C
Melting point: -114.2°C
Density (gas): 1.00045 g/l
Solubility in water, g/100ml at 30°C: 67 (moderate)
Relative vapour density (air = 1): 1.3
Octanol/water partition coefficient as log Pow: 0.25

EXPOSURE & HEALTH EFFECTS

Routes of exposure
Serious local effects by all routes of exposure. The substance can be absorbed into the body by inhalation.

Effects of short-term exposure
Rapid evaporation of the liquid may cause frostbite. The substance is corrosive to the eyes, skin and respiratory tract. Inhalation of this gas may cause asthma-like reactions (RADS). Exposure could cause asphyxiation due to swelling in the throat. Inhalation of high concentrations may cause lung oedema, but only after initial corrosive effects on the eyes and the upper respiratory tract have become manifest. Inhalation of high concentrations may cause pneumonitis. See Notes.

Inhalation risk
A harmful concentration of this gas in the air will be reached very quickly on loss of containment.

Effects of long-term or repeated exposure
Repeated or prolonged inhalation may cause effects on the teeth. This may result in tooth erosion. The substance may have effects on the upper respiratory tract and lungs. This may result in chronic inflammation of the respiratory tract and reduced lung function . Mists of this strong inorganic acid are carcinogenic to humans. See Notes.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 2 ppm as STEL; A4 (not classifiable as a human carcinogen).
MAK: 3.0 mg/m^3, 2 ppm; peak limitation category: I(2); pregnancy risk group: C.
EU-OEL: 8 mg/m^3, 5 ppm as TWA; 15 mg/m^3, 10 ppm as STEL

ENVIRONMENT

NOTES

The occupational exposure limit value should not be exceeded during any part of the working exposure.
The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.
IARC considers mists of strong inorganic acid to be carcinogenic (group 1). However there is no information available on the carcinogenicity of other physical forms of this substance. Therefore no classification for carcinogenicity under GHS has been applied.
Turn leaking cylinder with the leak up to prevent escape of gas in liquid state.
Other UN number(s) 2186 (refridgerated liquid) hazard class: 2.3; subsidiary hazard: 8; 1789 (hydrochloric acid) hazard class: 8, pack group II or III.
Aqueous solutions may contain up to 38% hydrogen chloride.

ADDITIONAL INFORMATION

EC Classification
Symbol: T, C; R: 23-35; S: (1/2)-9-26-36/37/39-45